

- **PROJECT TITLE**

Login and Registration System Using C++

- **Project Description**

The Login and Registration System is a C++ application that allows users to create an account and securely log in using a username and password. User credentials are stored in a file, enabling data persistence even after the program is closed. The system validates user input, prevents duplicate usernames, and verifies login credentials before granting access.

- **Objectives**

To create a user registration system.

To accept username and password from users.

To validate user inputs.

To prevent duplicate usernames.

To store user credentials in a file.

To verify user identity during login.

To display appropriate success and error messages.

Features

User Registration

User Login

File Handling

Username Validation

Password Verification

Error Handling

System Requirements

C++ Compiler (GCC, Turbo C++, Visual Studio, etc.)

Text File Storage

Basic Knowledge of File Handling

• Algorithm

Registration Process

1. Start the program.
2. Enter username and password.
3. Check whether the username already exists.
4. If the username exists, display an error message.
5. Otherwise, store the username and password in a file.
6. Display registration success message.

Login Process

1. Enter username and password.
2. Read stored credentials from the file.
3. Compare entered credentials with stored data.
4. If both match, display login successful.
5. Otherwise, display invalid username or password.
6. End the program.

Data Storage Method

The program stores user credentials in a text file. Each record contains:

Username

Password

Example:

manyatha password123

john abc123

rahul rahul@123

- **Output Messages**

Registration Success

Registration Successful!

Duplicate Username

Username already exists!

Please choose another username.

Login Success

Login Successful!

Welcome User.

Login Failure

Invalid Username or Password!

- **Applications**

College Management Systems

Student Portals

Banking Applications

Library Management Systems

Online Learning Platforms

Employee Management Systems

- **Advantages**

Simple and easy to use.

Stores data permanently using files.

Prevents duplicate usernames.

Provides secure user authentication.

Demonstrates file handling concepts in C++.

- **Conclusion**

The Login and Registration System is a useful application that manages user authentication efficiently. It allows users to register and log in securely while storing credentials in files. This project helps in understanding important C++ concepts such as functions, file handling, conditional statements, loops, and user authentication mechanisms.